

1. Administrative & Study Identification

Full Study Title: A Phase III Prospective, Multicenter, Randomized, Open-Label Trial of Acalabrutinib Plus Venetoclax Versus Venetoclax Plus Obinutuzumab in Previously Untreated Chronic Lymphocytic Leukemia or Small Lymphocytic Lymphoma **Study Title:** A Study of Acalabrutinib Plus Venetoclax Versus Venetoclax Plus Obinutuzumab in Previously Untreated Chronic Lymphocytic Leukemia or Small Lymphocytic Lymphoma **Short Title/Acronym:** MAJIC **Study Protocol Number:** D8220C00027 **NCT Number:** NCT05057494 **Posting ID:** 124559742270511639 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Sponsor / Company Name:** AstraZeneca **Investigational Product:** Acalabrutinib + Venetoclax vs. Venetoclax + Obinutuzumab (Gazyva) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess whether minimal residual disease (MRD) driven finite acalabrutinib plus venetoclax (AV) treatment is non-inferior to MRD-driven finite venetoclax plus obinutuzumab (VO) treatment with respect to progression-free survival (PFS). **Secondary Objectives:** To evaluate the rates of peripheral blood and bone marrow undetectable minimal residual disease (uMRD) via clonoSEQ® and flow cytometry, as well as overall survival (OS).

2.2 Background & Rationale While multiple effective frontline regimens exist for previously untreated Chronic Lymphocytic Leukemia (CLL) or Small Lymphocytic Lymphoma (SLL), it remains unclear which targeted doublet combination is optimal and what the appropriate duration of time-limited frontline therapy should be. The MAJIC trial aims to address these unanswered questions by directly comparing an all-oral, time-limited regimen (AV) against a standard time-limited chemoimmunotherapy-free regimen (VO), utilizing an individualized, MRD-guided treatment duration approach.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Venetoclax plus Obinutuzumab) **Blinding:** Open-label **Randomization:** Randomized (1:1) to Arm A (MRD-limited finite AV therapy) or Arm B (MRD-limited finite VO therapy)

3.2 Planned Enrollment & Duration Target **N**: 607 evaluable patients **Number of Sites**: Multicenter / International **Estimated Study Start**: October 2021 **Estimated Primary Completion**: November 30, 2029 **Individual Participant Involvement**: Approximately 5 years total duration

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of signing the consent form. 2. Documented treatment-naïve (TN) CLL/SLL requiring treatment according to iwCLL 2018 guidelines. 3. Adequate bone marrow (BM) function independent of growth factor or platelet transfusion support within 2 weeks of screening (ANC $\geq 1.0 \times 10^9/L$, or ≥ 500 cells/ μL with documented BM involvement; Platelets $\geq 30 \times 10^9/L$, or $\geq 10 \times 10^9/L$ with documented BM involvement). 4. Estimated Creatinine Clearance (CrCL) ≥ 30 mL/min calculated by Cockcroft-Gault. 5. Serum AST and ALT $\leq 3 \times$ ULN; Total bilirubin $\leq 1.5 \times$ ULN. 6. ECOG performance status of 0 to 2.

4.2 Exclusion Criteria 1. Clinically significant cardiovascular disease, such as symptomatic arrhythmias, congestive heart failure, or myocardial infarction within 6 months of screening. 2. Child-Pugh B/C liver cirrhosis. 3. Active infections including HIV, positive cytomegalovirus (CMV) DNA PCR, or serologic status reflecting active hepatitis B (HBV) or C (HCV) infection. 4. History of or ongoing confirmed progressive multifocal leukoencephalopathy (PML). 5. Prior CLL/SLL-specific therapies (except rituximab for autoimmune cytopenias), prior radio- or toxin-conjugated antibody therapy, or prior allogeneic/autologous stem cell transplant. 6. History of significant cerebrovascular disease/event (e.g., stroke or intracranial hemorrhage) within 6 months. 7. Corticosteroid use > 20 mg within 1 week before the first dose of study intervention. 8. Requires treatment with a strong CYP3A4 inhibitor/inducer, or requires/receiving anticoagulation with warfarin or equivalent vitamin K antagonists. 9. Requires treatment with proton pump inhibitors (PPIs) if receiving acalabrutinib capsules. 10. Vaccination with live vaccines within 28 days prior to screening, or major surgical procedure within 28 days of first dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A (AV):** Acalabrutinib 100 mg once daily + Venetoclax 40 mg once daily (subsequently ramping up to standard dose). * **Arm B (VO):** Venetoclax 400 mg once daily (following a 5-week ramp-up) + Obinutuzumab (Gazyva) 1000 mg on a daily specified schedule per 28-day cycles. **Route of Administration:** Oral (Acalabrutinib, Venetoclax) and Intravenous (Obinutuzumab). **Duration of**

Treatment: MRD-guided finite therapy. Treatment occurs in 28-day cycles. Therapy may be stopped after 12 cycles if undetectable MRD and at least a partial response (PR) are achieved. Otherwise, treatment continues for up to a maximum of 24 cycles.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care, non-warfarin anticoagulants, and H2-receptor antagonists or antacids for participants with gastric reflux. **Prohibited:** Strong CYP3A4 inhibitors (within 1 week of first dose) or strong CYP3A inducers (within 3 weeks of first dose), warfarin/vitamin K antagonists, and PPIs (applicable only to acalabrutinib capsule formulation).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Baseline Labs, Cardiac & Viral screening, Eligibility check
Baseline	Day 0	Randomization, First Dose, TLS risk assessment
Interim	Cycle 12	Safety Labs, Efficacy Assessment (iwCLL criteria), MRD testing to guide therapy duration
End of Study	Cycle 24	Final Treatment Assessment, Final MRD status, Transition to survival follow-up phase

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS). **Measurement Method:** Time from randomization to objective progressive disease per International Workshop on Chronic Lymphocytic Leukemia (iwCLL) 2018 criteria as assessed by the investigator, or death from any cause.

7.2 Secondary Endpoints 1. Rate of peripheral blood (PB) undetectable minimal residual disease (uMRD) based on a clonoSEQ® assay. 2. Rate of peripheral blood (PB) and Bone marrow (BM) undetectable minimal residual disease (uMRD) by flow cytometry. 3. Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits and hematology labs, with rigorous monitoring for Tumor Lysis Syndrome (TLS) during the venetoclax ramp-up phase, cytopenias, bleeding risk, and infections. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent safety monitoring committee overseeing trial safety globally.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcome measures, Safety Set for safety events. **Sample Size Justification:** \$N=607\$ powered specifically to test the non-inferiority of the experimental AV arm against the active comparator VO arm. **Handling of Missing Data:** Standard survival analysis algorithms utilizing event censoring for progression-free survival (PFS) and overall survival (OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Defined onsite and remote monitoring visits, focusing on source data verification of iwCLL responses and safety events. **Audit Trail:** Robust eCRF locking mechanisms and standardized data clarification forms.

11. Study Identifiers & Governance

Primary ID: D8220C00027 **Registry Number:** NCT05057494 **Posting ID:** 124559742270511639 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Study of Acalabrutinib Plus Venetoclax Versus Venetoclax Plus Obinutuzumab in Previously Untreated Chronic Lymphocytic Leukemia or Small Lymphocytic Lymphoma **Official Regulatory Title:** A Phase III Prospective, Multicenter, Randomized, Open-Label Trial of Acalabrutinib Plus Venetoclax Versus Venetoclax Plus Obinutuzumab in Previously Untreated Chronic Lymphocytic Leukemia or Small Lymphocytic Lymphoma

12. Clinical Framework (CLL/SLL Specific)

Disease Area: Chronic Lymphocytic Leukemia or Small Lymphocytic Lymphoma **Preferred UMLS Name:** B-cell chronic lymphocytic leukemia/small lymphocytic lymphoma **MeSH Code:** D007954 **SNOMED ID:** 13444005 **Diagnosis Threshold:** Treatment-naïve disease requiring treatment per iwCLL 2018 guidelines **Medical Methodology:** Minimal Residual Disease (MRD)-driven finite therapy **Optimization Target:** Undetectable Minimal Residual Disease (uMRD)

13. Study Procedures & Methodology

Management Pathway: Targeted doublet therapy administration (AV vs. VO combination pathways) **Education Protocol (HCP):** Venetoclax ramp-up schedules and Tumor Lysis Syndrome (TLS) mitigation protocols **Education Protocol (Patient):** Oral medication adherence instructions and infusion center scheduling **Quality Audit**

Mechanism: Centralized laboratory processing for deep molecular MRD testing via clonoSEQ® and flow cytometry.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 5 years of individual participant involvement, encompassing up to 24 cycles (~96 weeks) of active treatment, followed by extended survival and safety follow-up.

Onsite Visit Cadence: Regular intervals aligning with 28-day

cycles **Remote Monitoring Frequency:** Between treatment cycles where appropriate for oral therapy adherence **Checklist Review**

Interval: Cycle 12 and Cycle 24 (Key MRD evaluation timepoints)

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					